

VAISHNAVI SHARMA

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SUMMARY

AI/ML and full-stack developer with experience building PyTorch/FastAPI inference pipelines, React/Next.js interfaces, Supabase/PostgreSQL-backed apps, and open-source contributions. Interested in building analytics, automation, and AI-assisted workflows for institutional-scale systems.

EDUCATION

Delhi Technological University — B.Tech, Computer Science Engineering : CGPA: 8.2/10

New Delhi | 2025 – 2029

JLN International School — CBSE, Class XII: 92.8%

New Delhi | 2025

Modern Delhi International School — CBSE, Class X: 97.8%

Faridabad | 2023

TECHNICAL SKILLS

AI/ML & Data: Python, PyTorch, Scikit-learn, Pandas, NumPy, Librosa, OpenCV, Matplotlib, Seaborn, ML inference pipelines, data visualisation

Frontend: React, Next.js, TypeScript, JavaScript, Tailwind CSS, HTML/CSS

Backend & Data: FastAPI, Supabase (PostgreSQL), REST API design

Tools & Practices: Git, GitHub, Google Colab, Vercel, Hugging Face Spaces, C, C++

PROJECTS

DeepDetect: Audio Deepfake Detection | PyTorch, Wav2Vec2, FastAPI, Librosa, NumPy | [GitHub](#)

- Collaborated in a team to build an audio deepfake detection pipeline using Wav2Vec2 embeddings, Librosa preprocessing, Mel-spectrogram features, and FastAPI model-serving logic.
- Designed confidence scoring and risk-label outputs with acoustic evidence features to make model predictions interpretable for verification workflows.
- Integrated preprocessing, inference, and frontend display into an end-to-end prototype, reducing the output from a black-box score to a user-readable verification report.
- Ran a pilot evaluation on 20 curated test clips, achieving 82.4% accuracy and identifying failure cases for future dataset expansion.

MNIST Digit Recognizer | Scikit-learn, KNN, Gradio, Hugging Face Spaces | [GitHub](#)

- Built and deployed an interactive handwritten digit classifier with a browser-based drawing pad, preprocessing pipeline, trained KNN model, and prediction confidence display.
- Compared Logistic Regression, KNN, and Decision Tree baselines, with KNN achieving ~95% accuracy on the test split.
- Packaged the model using Joblib and deployed the app on Hugging Face Spaces with a documented GitHub repository.

Text Rendering in Diffusion Models | PyTorch, Diffusion Models, OCR Metrics, Computer Vision | Ongoing research initiative

- Working in a 3-member research initiative on glyph conditioning and localized glyph-region loss to improve legible text generation in diffusion models.
- Defined evaluation metrics for OCR readability, glyph accuracy, spacing, alignment, and region-level visual fidelity.
- Preparing baseline-vs-conditioned experiment design with OCR-based evaluation, qualitative failure analysis, and lightweight training objectives.

PACT | Habit Accountability Tracker | React, Supabase | Locally used application

- Built a two-user habit tracking web app with gamified progress mechanics and shared accountability features, run and tested locally.
- Designed the Supabase data schema and React component architecture to sync habit state and progress between two accountability partners in real time.
- Implemented persistent local state management for offline-first tracking between sessions.

Personal Portfolio Website | Next.js, TypeScript, Tailwind CSS, Framer Motion, GSAP | [GitHub](#)

- Built a responsive portfolio with animated interactions, project/skills/education sections, deployed pages, reusable components, and a photography gallery.
- Structured reusable UI components and smooth motion effects to present technical projects, achievements, and creative photography work in one deployed site.

EXPERIENCE

GirlScript Summer of Code - Open Source Contributor | Remote | 2026

- Selected as a GSSoC 2026 contributor; merged 5 pull requests through issue-based fixes and collaborative code reviews with maintainers.

POSITIONS OF RESPONSIBILITY

- **IEEE DTU Student Branch – EA/IR & Membership Development Executive** : led sponsor and vendor outreach for Vihaan 9.0 and Invictus 2026; secured 5+ partnerships, including ₹75K deal.
- **DTU Times – Photographer**: Covered 6 large-scale campus events as student photographer, contributing visual storytelling and photojournalism.
- **Cognitive Minds, DTU** – Debating Society Member: represented DTU at IIT Kanpur in the inter-collegiate competitive debating circuit.

ACHIEVEMENTS & AWARDS

- Best IEEE Award, Vihaan 9.0 (Apr 2026): recognized for technical innovation and team execution at a 40-hour hackathon with 10,000+ registrations.
- Smart India Hackathon - University Round: advanced to university-level selection in first year through competitive shortlisting.

CERTIFICATES

Machine Learning Specialization - Coursera, Andrew Ng / Stanford University | [Certificate](#)

Adobe AI Scholastics Program - Diffusion Models; 2-day hands-on PyTorch workshop, 2026